

ROS 2 Foundations

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#Empowering Minds.

1 ROS 2 Humble Foundations

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- Environment and First Observation
- The ROS 2 Graph and Nodes
- Topics, Publishers, Subscribers, and Messages
- Services and Actions
- Parameters, Packages, and Workspaces
- Python Publisher and Subscriber
- Drone-System Integration
- Troubleshooting and Practice

From Python to ROS 2

Python provides logic; ROS 2 connects that logic to the robotic system

A drone is not one program. It is a team of communicating components.

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Perception

Camera acquisition and object detection.

Mission Logic

Trajectory generation and inspection decisions.

Flight Interface

Telemetry reception and PX4 setpoint publication.

ROS 2 Role

ROS 2 provides the communication and integration framework that allows these components to discover one another and exchange typed information.

What Is ROS 2?

A modular software framework for building robot systems

Intuitive Definition

ROS 2 is a collection of software libraries, command-line tools, conventions, and communication mechanisms used to create and integrate robotic applications.

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Team Analogy

Imagine a technical team in which each specialist has one responsibility. ROS 2 provides the shared language, communication channels, addresses, and rules that let the specialists cooperate.

Important Distinction

ROS 2 is not the computer's operating system. Ubuntu manages hardware, files, memory, users, and processes. ROS 2 runs on top of Ubuntu and organizes communication between robotics software components.

Why Use ROS 2 in Robotics?

Complex systems become understandable when responsibilities are separated

- ▶ **Modularity:** camera, mission, control, and logging components can be developed separately.
- ▶ **Typed communication:** interfaces define the structure of exchanged data.
- ▶ **Observability:** terminal tools expose nodes, topics, services, actions, and parameters.
- ▶ **Language flexibility:** components may be implemented in Python or C++.
- ▶ **Reuse:** packages and standard interfaces reduce duplicated engineering work.
- ▶ **Distributed operation:** communicating components may run in different processes or on different computers.

Drone Project Connection

ROS 2 connects mission logic, PX4 interfaces, camera bridges, perception programs, visualization, and evidence collection without forcing them into one monolithic program.

Course Compatibility Contract

Every command and exercise targets one supported baseline

Ubuntu 22.04 LTS + ROS 2 Humble Hawksbill

Student Rule

Use the versions specified by the internship repository. Do not silently replace Humble commands or packages with instructions written for another ROS 2 distribution.

Why This Matters

Package names, supported platforms, simulator combinations, APIs, and setup instructions may differ between ROS 2 distributions. Newer does not automatically mean compatible with this project.

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Live Check 1: Load ROS 2 Humble

Each new terminal must receive the ROS 2 environment

Execute

```
source /opt/ros/humble/setup.bash
printenv ROS_DISTRO
which ros2
ros2 --help
```

Expected Evidence

ROS_DISTRO should report humble, and which ros2 should locate the ROS 2 command-line executable.

Interpretation

source changes the current shell environment. It does not reinstall ROS 2. Repeat it in every new terminal unless the project defines an approved automatic setup.

Live Check 2: Diagnose the Environment

Verify the installation before blaming application code

Execute

```
ros2 doctor --report
```

What to Inspect

Confirm the ROS 2 distribution, platform, middleware information, network interfaces, and package environment. Read warnings in context rather than treating every line as a fatal error.

Engineering Habit

Preserve the command and complete output when asking for help. A screenshot of only the last line may hide the actual cause.

Live Demonstration: Start Turtlesim

Use a small system before connecting the drone stack

Terminal 1

```
source /opt/ros/humble/setup.bash  
ros2 run turtlesim turtlesim_node
```

Terminal 2

```
source /opt/ros/humble/setup.bash  
ros2 run turtlesim turtle_teleop_key
```

What to Observe

Two executables run as separate processes. One displays the simulated turtle; the other converts keyboard input into movement commands.

Terminal Focus

Keep Terminal 2 active while pressing arrow keys. Stop each foreground process deliberately with `Ctrl+C`.

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The ROS 2 Graph

A live map of components and communication relationships

Nodes + topics + services + actions + parameters = **ROS 2 graph**

Intuition

The graph is not a file stored once. It is the changing network of ROS 2 entities that currently exist and communicate while the system runs.

Drone Example

An orbit controller, PX4 bridge, image bridge, detector, and evidence logger appear as nodes connected through named interfaces.

What Is a Node?

One focused participant in the robotic software system

Definition

A node is a ROS 2 participant responsible for a focused purpose. It may publish, subscribe, provide services, execute actions, use parameters, and write logs.

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Specialist Analogy

A camera node produces images. A detector node interprets them. A mission node creates trajectory references. A logger node preserves evidence. Each specialist has a clear responsibility.

Design Principle

Avoid one giant node that performs every task. Clear responsibilities make components easier to test, replace, diagnose, and reuse.

Live Tutorial: Discover Nodes

Ask the running system who is participating

Keep Turtlesim Running, Then Execute

```
ros2 node list  
ros2 node info /turtlesim  
ros2 node info /teleop_turtle
```

Read the Output

`ros2 node list` identifies active node names. `ros2 node info` reveals publishers, subscribers, services, actions, and interfaces associated with one node.

Project Transfer

Later, use `ros2 node list` to verify that the orbit controller, bridges, and perception nodes actually started. A terminal window alone is not sufficient evidence.

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Publishers send data; subscribers receive matching data

Publisher $\longrightarrow$ named topic $\longrightarrow$ **Subscriber**

A topic resembles a radio channel. Publishers transmit without addressing one specific receiver. Any compatible subscriber may listen to the named channel.

Camera images, vehicle odometry, desired paths, detection results, and telemetry are natural topic data because they may be produced repeatedly over time.

Publishers and Subscribers

Communication roles are independent from node names

Publisher

Creates messages of a defined type and sends them on a named topic.

Subscriber

Registers interest in a named topic and executes a callback when compatible data arrives.

Many-to-Many Possibility

One publisher may serve several subscribers, and several publishers may write compatible messages to the same topic. System design must make this intentional.

Messages: Typed Data Contracts

Every topic carries a defined data structure

Intuition

A message type is a standardized form. It defines the fields and field types that communicating components agree to exchange.

Examples

A velocity command may contain linear and angular components. An image message includes pixel data and metadata. PX4 odometry includes estimated position, velocity, orientation, and timestamps.

Why Types Matter

A subscriber cannot safely interpret arbitrary bytes. Both sides must agree on the interface, units, coordinate conventions, and semantic meaning.

Live Tutorial: Inspect Topics

Discover names, types, rates, and current data

```
ros2 topic list
ros2 topic list -t
ros2 topic info /turtle1/cmd_vel --verbose
ros2 topic type /turtle1/cmd_vel
ros2 interface show geometry_msgs/msg/Twist
ros2 topic echo /turtle1/pose
ros2 topic hz /turtle1/pose
```

Observation Sequence

Find the topic, identify its type, inspect the interface fields, observe actual messages, and measure publication frequency.

Stop Streaming Commands

`ros2 topic echo` and `ros2 topic hz` continue until interrupted. Stop them with `Ctrl+C`.

Send a typed velocity command from the terminal

What Happens

Flight-Safety Boundary

Terminal publication is appropriate for controlled tutorials and simulation. Never publish experimental commands to a real vehicle without an approved safety process and operator control.

Quality of Service: Communication Policy

Not every robotics data stream needs the same delivery behavior

Intuition

Quality of Service, or QoS, describes policies such as reliability, history depth, durability, and whether fresh data is preferred over recovering every old sample.

Drone Example

For high-rate sensor estimates, the newest value may be more useful than a delayed queue of old values. Configuration or state information may have different delivery needs.

Compatibility

A publisher and subscriber can use the same topic type and still fail to communicate if their QoS policies are incompatible. Inspect interfaces with `-verbose` when data is missing.

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Services: Request and Response

Ask for one operation and receive one result

Client request → **Service server** → **response**

Intuition

A service resembles asking a technician one question and waiting for one answer. It is best for short operations that complete quickly.

Drone Examples

Start an inspection mission, reset a component, query one configuration state, or request a controlled operation with an immediate response.

Not for Continuous Data

Use topics for streams. Do not repeatedly call a service to imitate high-rate telemetry.

Live Tutorial: Discover and Call a Service

Create a second turtle through a request-response interaction

```
ros2 service list
ros2 service type /spawn
ros2 interface show turtlesim/srv/Spawn
ros2 service call /spawn turtlesim/srv/Spawn \
"{x: 2.0, y: 2.0, theta: 0.0, name: 'inspector'}"
```

Read the Interaction

The service type defines request fields and response fields. The client sends position, orientation, and name; the server returns the created turtle name.

Common Mistake

A correct service name with an incorrect service type or malformed field structure will fail. Inspect the interface before composing the request.

Actions: Long-Running Goals

A goal may need feedback, time, completion, or cancellation

Goal $\longrightarrow$ repeated feedback $\longrightarrow$ result
with the possibility of cancellation

Intuition

An action resembles assigning a technician a task that takes time. You receive progress updates, can cancel if needed, and receive a final result.

Drone Examples

Execute a planned inspection segment, navigate to a waypoint, or perform a long-running scan where progress and cancellation matter.

Live Tutorial: Inspect and Send an Action Goal

Rotate the turtle while observing feedback

```
ros2 action list -t
ros2 action info /turtle1/rotate_absolute
ros2 interface show turtlesim/action/RotateAbsolute
ros2 action send_goal --feedback \
/turtle1/rotate_absolute \
turtlesim/action/RotateAbsolute "{theta: 1.57}"
```

What to Observe

The action client sends a target angle, receives periodic feedback while rotation continues, and finally receives a result.

Concept Check

If an operation is immediate, a service may fit. If it takes time and needs feedback or cancellation, an action is usually the clearer model.

Topics, Services, or Actions?

Choose the interface from the communication need

- ▶ **Topic:** repeated stream with no direct response, such as odometry or images.
- ▶ **Service:** short request with one response, such as resetting or starting a simple operation.
- ▶ **Action:** long-running goal with feedback, result, and cancellation, such as executing a mission segment.

Design Question

Do not begin by asking, “Which ROS 2 feature do I want to use?” Ask, “What communication behavior does this engineering interaction require?”

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Parameters: Node Configuration

Adjust behavior without rewriting source code

Intuition

Parameters are named configuration values associated with a node. They may express limits, rates, target names, radii, durations, or feature choices.

Drone Example

The orbit controller may expose `radius_m`, `altitude_m`, `orbit_period_s`, and `face_center` as parameters.

Validation

A configurable value still requires type, range, unit, and consistency checks before it influences vehicle behavior.

Live Tutorial: Inspect Parameters

Discover and modify node configuration

```
ros2 param list /turtlesim
ros2 param describe /turtlesim background_r
ros2 param get /turtlesim background_r
ros2 param set /turtlesim background_r 80
ros2 service call /clear std_srvs/srv/Empty "{}"
```

Observation

The parameter value changes the background configuration. Calling `/clear` redraws the window so the new value becomes visible.

Project Transfer

Use `ros2 param list /orbit_controller` to discover mission parameters instead of searching blindly through source code.

Packages: Deliverable Units of ROS 2 Software

Group code, metadata, configuration, and launch resources

Definition

A ROS 2 package is an organizational and build unit containing related executables, libraries, interfaces, tests, configuration, launch files, and dependency metadata.

Drone Examples

A mission package may contain the orbit controller and trajectory utilities. A perception package may contain camera-processing nodes and detector configuration.

Package Responsibility

A package should communicate a coherent purpose. Do not place every internship file into one undefined package.

Live Tutorial: Inspect Installed Packages

Use the package index rather than guessing locations

```
ros2 pkg list | grep turtlesim  
ros2 pkg executables turtlesim  
ros2 pkg prefix turtlesim  
ros2 interface package turtlesim
```

Interpretation

These commands confirm that the package is discoverable, list its executables, identify its installation prefix, and display its interfaces.

Environment Reminder

If a newly built package is missing, verify that the correct workspace was built and that its `install/setup.bash` file was sourced in the current terminal.

Workspaces: Where Packages Are Developed Together

A controlled environment for source, build products, and installation results

src/ $\longrightarrow$ build/ + install/ + log/

- ▶ `src/`: package source code and maintained project files.
- ▶ `build/`: intermediate files created during compilation or installation.
- ▶ `install/`: discoverable executables, libraries, and environment setup.
- ▶ `log/`: build and test logs used for diagnosis.

Mental Model

The workspace is the engineering workshop; packages are organized projects inside it; `colcon` performs the coordinated build.

Live Tutorial: Create a Humble Python Workspace

Build in a separate practice area before modifying the internship packages

```
source /opt/ros/humble/setup.bash
mkdir -p ~/ros2_humble_practice/src
cd ~/ros2_humble_practice/src

ros2 pkg create --build-type ament_python \
--license Apache-2.0 \
--node-name telemetry_demo robotics_practice
```

What Is Created

A Python ROS 2 package containing metadata, Python package structure, resource index entry, setup files, and a starter node.

Naming

Use lowercase package names with underscores. Do not create the practice package inside the installed ROS directory or an unrelated Git repository.

Live Tutorial: Build and Source the Workspace

Source code becomes a discoverable ROS 2 package

```
cd ~/ros2_humble_practice  
colcon build --symlink-install  
source install/setup.bash
```

```
ros2 pkg executables robotics_practice  
ros2 run robotics_practice telemetry_demo
```

Why `-symlink-install`?

For Python development, editable source files are linked into the install space, reducing unnecessary rebuilds for many code-only changes. Metadata changes may still require rebuilding.

Overlay Order

Source the base Humble installation first, then the workspace overlay. A new terminal forgets both until they are sourced again.

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Python Publisher: The Communication Pattern

A timer creates and publishes typed telemetry

```
import rclpy
from rclpy.node import Node
from std_msgs.msg import Float32

class AltitudePublisher(Node):
    def __init__(self):
        super().__init__('altitude_publisher')
        self.publisher = self.create_publisher(
            Float32, '/demo/altitude_m', 10)
        self.timer = self.create_timer(1.0, self.publish_altitude)

    def publish_altitude(self):
        msg = Float32()
        msg.data = 4.0
        self.publisher.publish(msg)
```

Read It as a Story

Create a node, create a typed publisher, create a periodic timer, build one message, and publish it on a named topic.

Python Subscriber: The Callback Pattern

Incoming telemetry triggers a focused method

```
import rclpy
from rclpy.node import Node
from std_msgs.msg import Float32

class AltitudeSubscriber(Node):
    def __init__(self):
        super().__init__('altitude_subscriber')
        self.subscription = self.create_subscription(
            Float32, '/demo/altitude_m', self.callback, 10)

    def callback(self, msg):
        self.get_logger().info(
            f'Received altitude: {msg.data:.2f} m')
```

Read It as a Story

Create a node, subscribe to a typed topic, and execute the callback whenever a compatible message arrives.

Live Tutorial: Verify a Publisher Before Integration

Use terminal tools as an independent observer

```
ros2 node list  
ros2 topic list -t  
ros2 topic info /demo/altitude_m --verbose  
ros2 topic echo /demo/altitude_m  
ros2 topic hz /demo/altitude_m
```

Layer-by-Layer Evidence

First confirm the node exists, then the topic and type, then endpoint compatibility, then actual values, and finally the observed rate.

Debugging Principle

Do not connect five unverified components and then ask why the pipeline fails. Verify each interface independently before integration.

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ROS 2 in the Autonomous-Drone Pipeline

Communication joins mission, flight, simulation, and perception

Command Flow

ROS 2 mission node → PX4 input topics → PX4 control → Gazebo motion

Evidence Flow

Gazebo camera → image bridge → ROS 2 image topic → detector → results and logs

Responsibility Boundary

ROS 2 transports and integrates information. PX4 performs flight estimation and control. Gazebo simulates physics. The AI detector produces observations, not direct safety-certified commands.

Live Tutorial: Inspect the Drone Graph

Use the same discovery method on the full project

```
source /opt/ros/humble/setup.bash
cd ~/quard_um6p_intership
source config/project.env
# After building the workspace:
source install/setup.bash
```

```
ros2 node list
ros2 topic list -t
ros2 service list -t
ros2 action list -t
ros2 param list
```

Question for Every Interface

Who produces it? Who consumes it? What type does it use? At what rate? In which units and coordinate frame? What observation proves it works?

Live Tutorial: Inspect PX4 Interfaces

Understand direction before publishing or subscribing

```
ros2 topic list | grep /fmu
ros2 topic info /fmu/out/vehicle_odometry --verbose
ros2 interface show px4_msgs/msg/VehicleOdometry
ros2 topic echo /fmu/out/vehicle_odometry --once

ros2 topic info /fmu/in/trajectory_setpoint --verbose
ros2 interface show px4_msgs/msg/TrajectorySetpoint
```

Naming Intuition

For PX4 interfaces, /fmu/out/... carries data leaving PX4, while /fmu/in/... carries data entering PX4.

No Blind Publication

Inspecting a setpoint interface is safe. Publishing arbitrary flight setpoints is a separate controlled activity performed only in the documented simulation workflow.

Visualize the Graph with `rqt_graph`

A diagram complements terminal evidence

Execute While Nodes Are Running

```
rqt_graph
```

What to Trace

Locate the mission node, PX4 communication path, camera bridge, perception node, and topics connecting them. Compare the graph with the intended architecture.

Interpretation

A visible connection shows discovered ROS 2 endpoints. It does not prove correct values, rates, units, QoS, or system behavior. Use CLI inspection and observed outputs as additional evidence.

Record and Replay Topic Data

Preserve communication evidence for repeatable analysis

```
ros2 bag record /turtle1/pose
# Move the turtle, then stop recording with Ctrl+C.
ros2 bag info rosbag2_*
ros2 bag play rosbag2_*
```

Intuition

A ROS bag records serialized topic messages with timing information. It allows later inspection or replay without reproducing the original motion immediately.

Data Management

High-rate images and telemetry can create large files quickly. Record only required topics, monitor storage, and keep generated bag data out of normal Git history unless the project explicitly requires a small fixture.

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A ROS 2 Troubleshooting Ladder

Move from environment to behavior without skipping layers

1. **Environment:** Was Humble sourced in this terminal?
2. **Process:** Is the expected executable still running?
3. **Node:** Does it appear in `ros2 node list`?
4. **Interface:** Does the topic, service, or action exist with the expected type?
5. **Compatibility:** Do endpoints and QoS policies match?
6. **Data:** Are values, rate, timestamp, units, and frame correct?
7. **Behavior:** Does the complete component produce the intended result?

Method

Change one factor at a time and preserve the original error. Random restarts may remove the symptom without explaining the cause.

Common Humble Problems and First Checks

Use observable evidence before reinstalling software

```
# ros2: command not found
source /opt/ros/humble/setup.bash

# Package not found after building
source ~/your_workspace/install/setup.bash

# Node expected but absent
ps aux | grep process_name
ros2 node list

# Topic exists but no data
ros2 topic info /topic_name --verbose
ros2 topic echo /topic_name

# Broad environment report
ros2 doctor --report
```

Do Not Guess

Confirm the Ubuntu release, ROS distribution, current workspace, active process, interface type, and complete terminal output before changing the installation.

Guided Exercise: Communication Detective

Explain a running ROS 2 system without reading its source code first

Mission

With Turtlesim running, identify both nodes, all velocity-related topics, the command message type, one service, one action, and one configurable parameter.

Required Evidence

Record the commands used, selected output, a simple hand-drawn graph, and one sentence explaining why each interface is a topic, service, action, or parameter.

Learning Goal

The objective is not command memorization. It is the ability to reconstruct the software architecture from observable ROS 2 interfaces.

Guided Exercise: Drone Interface Map

Transfer the same reasoning to the internship repository

Create a Small Interface Table in Your Notes

For one mission node and one perception node, record the node name, input interfaces, output interfaces, message types, expected rates, units, coordinate frames, and verification commands.

Minimum Questions

What information enters the component? What does it produce? Which ROS 2 mechanism is used? Who is connected? What evidence confirms meaningful data rather than only discovery?

Do Not Control the Vehicle Yet

This exercise is architecture inspection. Flight-command publication belongs to the later documented PX4 simulation tutorial.

ROS 2 Readiness Check

Students should now be able to explain and demonstrate

Conceptual Understanding

- ▶ ROS 2 versus Ubuntu
- ▶ graph, nodes, and interfaces
- ▶ publishers, subscribers, and messages
- ▶ topics, services, and actions
- ▶ packages, workspaces, and parameters

Practical Capability

- ▶ source ROS 2 Humble
- ▶ run and inspect nodes
- ▶ inspect and publish topic data
- ▶ call a service and send an action goal
- ▶ create, build, source, and run a package
- ▶ diagnose a missing interface systematically

Learning Roadmap

From communication foundations to the complete autonomous-drone system

ROS 2 graph → Python nodes → PX4 interfaces → Gazebo sensors → AI perception
→ mission evidence

Next Technical Step

Use these communication concepts to understand the project's Python nodes, PX4 message interfaces, image bridge, and launch sequence before integrating the full simulation pipeline.

Final Principle

A connected graph is only the beginning. A professional engineer also verifies interface types, QoS, data values, units, coordinate frames, timing, system behavior, and documented limitations.

Official ROS 2 Humble Resources I

Follow the Humble documentation for version-compatible tutorials



Open Robotics, *ROS 2 Humble Documentation*.

<https://docs.ros.org/en/humble/>



Open Robotics, *ROS 2 Humble Tutorials*.

<https://docs.ros.org/en/humble/Tutorials.html>



Open Robotics, *Beginner CLI Tools*.

<https://docs.ros.org/en/humble/Tutorials/Beginner-CLI-Tools.html>



Open Robotics, *ROS 2 Concepts*.

<https://docs.ros.org/en/humble/Concepts.html>

Official ROS 2 Humble Resources II

Packages, Python nodes, and integration practice



Open Robotics, *Creating a Workspace*.

<https://docs.ros.org/en/humble/Tutorials/Beginner-Client-Libraries/Creating-A-Workspace/Creating-A-Workspace.html>



Open Robotics, *Creating a Package*.

<https://docs.ros.org/en/humble/Tutorials/Beginner-Client-Libraries/Creating-Your-First-ROS2-Package.html>



Open Robotics, *Writing a Simple Publisher and Subscriber in Python*.

<https://docs.ros.org/en/humble/Tutorials/Beginner-Client-Libraries/Writing-A-Simple-Py-Publisher-And-Subscriber.html>



Open Robotics, *Writing a Simple Service and Client in Python*.

<https://docs.ros.org/en/humble/Tutorials/Beginner-Client-Libraries/Writing-A-Simple-Py-Service-And-Client.html>